THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Quadrilaterals** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 90°	The angles of a quadrilateral add up to 360°
2	**(d)** 115°	Adjacent angles of a parallelogram add up to 180°
3	**(b)** 58°	Opposite angles of a parallelogram are equal
4	**(a)** 24 cm	$4 \times 6 = 24$
5	**(c)** 26 cm	$2(8 + 5)$
6	**(a)** 10 cm	Diagonals of a rectangle are equal
7	**(a)** 8 cm	Diagonals of a parallelogram bisect each other
8	**(d)** 4 sides and 4 vertices	Quad means four
9	**(d)** 100°	The angles of a quadrilateral add up to 360°
10	**(a)** 108°	Adjacent angles of a parallelogram add up to 180°

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Sum of angles $= 360^\circ$ (1); fourth angle $= 360 - 270 = 90^\circ$ (1).	1+1
12	Opposite angle 48° (1); adjacent angles $180 - 48 = 132^\circ$ each (1).	1+1
13	Square (1); diagonals are equal and perpendicular (1).	1+1
14	Sum of angles $= 360^\circ$ (1); fourth angle $= 360 - 240 = 120^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\angle B = 180 - 62 = 118^\circ$ (1); $\angle C = \angle A = 62^\circ$ (1); $\angle D = \angle B = 118^\circ$ (1).	1+1+1
16	BD = AC = 16 cm (1); OA = half of AC = 8 cm (1); OB = 8 cm (1).	1+1+1
17	Diagonals bisect each other (1); AC = 2 x 5 = 10 cm (1); BD = 2 x 8 = 16 cm (1).	1+1+1
	OR: $\angle B = 180 - 78 = 102^\circ$ (1); $\angle C = \angle A = 78^\circ$ (1); $\angle D = \angle B = 102^\circ$ (1).	1+1+1
18	BD = AC = 20 cm (1); OA = half of AC = 10 cm (1); OB = 10 cm (1).	1+1+1
	OR: Diagonals bisect each other (1); AC = 2 x 6 = 12 cm (1); BD = 2 x 9 = 18 cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Fourth angle $= 95^\circ$ (2); exterior angles $180 - 70 = 110^\circ, 70^\circ, 95^\circ, 85^\circ$ (2); their sum $= 360^\circ$ (1).	2+2+1
	OR: $\angle B = 114^\circ$ (1); $\angle C = \angle A = 66^\circ$ (1); $\angle D = \angle B = 114^\circ$ (1); perimeter $= 4 \times 7 = 28$ cm (2).	1+1+1+2
20	Fourth angle $= 80^\circ$ (2); exterior angles $180 - 95 = 85^\circ, 100^\circ, 75^\circ, 100^\circ$ (2); their sum $= 360^\circ$ (1).	2+2+1
	OR: $\angle B = 106^\circ$ (1); $\angle C = \angle A = 74^\circ$ (1); $\angle D = \angle B = 106^\circ$ (1); perimeter $= 4 \times 9 = 36$ cm (2).	1+1+1+2

Mistakes to avoid:

- Saying the diagonals of a rhombus are equal, or the diagonals of a rectangle are perpendicular; it is the other way round.
- Treating adjacent angles of a parallelogram as equal; **opposite** angles are equal and adjacent angles add up to 180°.
- Forgetting that every square is a rectangle and a rhombus, and every rectangle or rhombus is a parallelogram.
- Using the sum 180° (triangle) instead of 360° for the interior angles of a quadrilateral.
- Saying a trapezium has two pairs of parallel sides; only one pair is parallel (two pairs make a parallelogram).
- Assuming a quadrilateral with equal diagonals must be a rectangle; the diagonals must also bisect each other, and a parallelogram is needed first.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Quadrilaterals** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 76°	Opposite angles of a parallelogram are equal
2	**(d)** 36 cm	$4 \times 9 = 36$
3	**(b)** 38 cm	$2(12 + 7)$
4	**(d)** 14 cm	Diagonals of a rectangle are equal
5	**(b)** 14 cm	Diagonals of a parallelogram bisect each other
6	**(a)** 90°	A rectangle has four right angles
7	**(b)** 95°	The angles of a quadrilateral add up to 360°
8	**(a)** 72°	Adjacent angles of a parallelogram add up to 180°
9	**(c)** 103°	Opposite angles of a parallelogram are equal
10	**(c)** 44 cm	$4 \times 11 = 44$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Opposite angle 82° (1); adjacent angles $180 - 82 = 98^\circ$ each (1).	1+1
12	360° (1); a diagonal splits it into two triangles, each with sum 180° (1).	1+1
13	Sum of angles $= 360^\circ$ (1); fourth angle $= 360 - 260 = 100^\circ$ (1).	1+1
14	Opposite angle 100° (1); adjacent angles $180 - 100 = 80^\circ$ each (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\angle B = 180 - 94 = 86^\circ$ (1); $\angle C = \angle A = 94^\circ$ (1); $\angle D = \angle B = 86^\circ$ (1).	1+1+1
16	BD = AC = 24 cm (1); OA = half of AC = 12 cm (1); OB = 12 cm (1).	1+1+1
17	Diagonals bisect each other (1); AC = 2 x 7 = 14 cm (1); BD = 2 x 11 = 22 cm (1).	1+1+1
	OR: $\angle B = 180 - 106 = 74^\circ$ (1); $\angle C = \angle A = 106^\circ$ (1); $\angle D = \angle B = 74^\circ$ (1).	1+1+1
18	BD = AC = 30 cm (1); OA = half of AC = 15 cm (1); OB = 15 cm (1).	1+1+1
	OR: Diagonals bisect each other (1); AC = 2 x 9 = 18 cm (1); BD = 2 x 13 = 26 cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Fourth angle $= 80^\circ$ (2); exterior angles $180 - 60 = 120^\circ, 80^\circ, 60^\circ, 100^\circ$ (2); their sum $= 360^\circ$ (1).	2+2+1
	OR: $\angle B = 124^\circ$ (1); $\angle C = \angle A = 56^\circ$ (1); $\angle D = \angle B = 124^\circ$ (1); perimeter $= 4 \times 12 = 48$ cm (2).	1+1+1+2
20	Fourth angle $= 105^\circ$ (2); exterior angles $180 - 88 = 92^\circ, 88^\circ, 105^\circ, 75^\circ$ (2); their sum $= 360^\circ$ (1).	2+2+1
	OR: $\angle B = 68^\circ$ (1); $\angle C = \angle A = 112^\circ$ (1); $\angle D = \angle B = 68^\circ$ (1); perimeter $= 4 \times 15 = 60$ cm (2).	1+1+1+2

Mistakes to avoid:

- Saying the diagonals of a rhombus are equal, or the diagonals of a rectangle are perpendicular; it is the other way round.
- Treating adjacent angles of a parallelogram as equal; **opposite** angles are equal and adjacent angles add up to 180°.
- Forgetting that every square is a rectangle and a rhombus, and every rectangle or rhombus is a parallelogram.
- Using the sum 180° (triangle) instead of 360° for the interior angles of a quadrilateral.
- Saying a trapezium has two pairs of parallel sides; only one pair is parallel (two pairs make a parallelogram).
- Assuming a quadrilateral with equal diagonals must be a rectangle; the diagonals must also bisect each other, and a parallelogram is needed first.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Quadrilaterals** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 32 cm	$2(10+6)$
2	**(d)** 18 cm	Diagonals of a rectangle are equal
3	**(b)** 18 cm	Diagonals of a parallelogram bisect each other
4	**(b)** parallel	Definition of a parallelogram
5	**(c)** 100°	The angles of a quadrilateral add up to 360°
6	**(a)** 65°	Adjacent angles of a parallelogram add up to 180°
7	**(b)** 125°	Opposite angles of a parallelogram are equal
8	**(d)** 56 cm	$4 \times 14 = 56$
9	**(a)** 48 cm	$2(15+9)$
10	**(d)** 26 cm	Diagonals of a rectangle are equal

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Trapezium (1); e.g. the side view of a bucket or a tabletop shape (1).	1+1
12	Sum of angles $= 360^\circ$ (1); fourth angle $= 360 - 270 = 90^\circ$ (1).	1+1
13	Opposite angle 124° (1); adjacent angles $180 - 124 = 56^\circ$ each (1).	1+1
14	They are equal (1); they bisect each other (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\angle B = 180 - 118 = 62^\circ$ (1); $\angle C = \angle A = 118^\circ$ (1); $\angle D = \angle B = 62^\circ$ (1).	1+1+1
16	BD = AC = 36 cm (1); OA = half of AC = 18 cm (1); OB = 18 cm (1).	1+1+1
17	Diagonals bisect each other (1); AC = 2 x 10 = 20 cm (1); BD = 2 x 12 = 24 cm (1).	1+1+1
	OR: $\angle B = 180 - 70 = 110^\circ$ (1); $\angle C = \angle A = 70^\circ$ (1); $\angle D = \angle B = 110^\circ$ (1).	1+1+1
18	BD = AC = 42 cm (1); OA = half of AC = 21 cm (1); OB = 21 cm (1).	1+1+1
	OR: Diagonals bisect each other (1); AC = 2 x 8 = 16 cm (1); BD = 2 x 15 = 30 cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Fourth angle $= 90^\circ$ (2); exterior angles $180 - 115 = 65^\circ, 115^\circ, 90^\circ, 90^\circ$ (2); their sum $= 360^\circ$ (1).	2+2+1
	OR: $\angle B = 96^\circ$ (1); $\angle C = \angle A = 84^\circ$ (1); $\angle D = \angle B = 96^\circ$ (1); perimeter $= 4 \times 8 = 32$ cm (2).	1+1+1+2
20	Fourth angle $= 96^\circ$ (2); exterior angles $180 - 72 = 108^\circ, 72^\circ, 96^\circ, 84^\circ$ (2); their sum $= 360^\circ$ (1).	2+2+1
	OR: $\angle B = 82^\circ$ (1); $\angle C = \angle A = 98^\circ$ (1); $\angle D = \angle B = 82^\circ$ (1); perimeter $= 4 \times 10 = 40$ cm (2).	1+1+1+2

Mistakes to avoid:

- Saying the diagonals of a rhombus are equal, or the diagonals of a rectangle are perpendicular; it is the other way round.
- Treating adjacent angles of a parallelogram as equal; **opposite** angles are equal and adjacent angles add up to 180°.
- Forgetting that every square is a rectangle and a rhombus, and every rectangle or rhombus is a parallelogram.
- Using the sum 180° (triangle) instead of 360° for the interior angles of a quadrilateral.
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